

Simon Zhu

University of Waterloo Mechatronics
Undergraduate

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Mechatronics Engineering student with a strong hands-on background in robotics, mechanical design, prototyping, embedded systems, and controls. Experienced in independently designing, fabricating, assembling, testing, and troubleshooting complete electromechanical systems, fluent in English, French and Mandarin.

Projects

Check-Mate: Autonomous Chess Board | C++, Arduino, Fusion 360

Dec.2023 – Mar.2025

- Provincial finalist, Hydro-Québec Science Fair: OIQ award for technical complexity, OCTAS Youth Innovation award, and a \$2,000 Université de Sherbrooke scholarship with a self-playing chess board.
- Integrated 64 linear Hall-effect sensors through four analog multiplexers, reducing Arduino input requirements from 64 to 4 while detecting chess-piece color through magnet polarity.
- Designed and fabricated a custom 4-MOSFET H-bridge after commercial drivers failed under electromagnet back EMF.
- Fit the CoreXY gantry within 2 mm of the playing area, with collision-checked paths and thermistor-triggered cooling.

Wheel-Leg Balancing Biped | C++, ESP32, SolidWorks, RL Training, Inverse Kinematics

Aug.2026 – Present

- Designing a 1 kg, 6-motor wheel-leg robot with 5-bar legs on 3D-printed 4.2:1 planetary gearboxes, split across two ESP32-S3s and with field-oriented control, a magnetic encoder, and an IMU with a Kalman filter.
- Trained a PPO trained neural-network controller with reinforcement learning in a physics simulation and used the simulation to choose the gear ratio and limb sizes before printing parts.

Swerve Drive | C++, Arduino, Closed Loop Control, Iterative Design

Sep.2025 – Mar.2026

- Independently designed and fabricated a fully 3D-printed coaxial swerve drive over 5 design iterations for approximately \$50 per module, down from ~\$400 commercial options, each swappable in 30 seconds with 4 bolts.
- Closed the steering loop with optical encoders on continuous-rotation servos, keeping all modules synchronized while driving.

Leadership & Teamwork Experience

John Abbott Robotics Club President | Robot Design Lead

Montréal, QC | Oct.2025 – Jun.2026

- Led a 30+ member club to **2nd place for Robot design Award** at the Canadian Robotics Competition, managing a budget of \$6,000, project timelines, submission deadlines and task allocation.
- Taught weekly workshops in CAD (Fusion 360), C++ and manufacturing, committing 6 hours/week.

Volunteering | Hospital Aide - St-Mary's Hospital & Elderly Activity Volunteer - Nova

Montréal, QC | Oct.2024 – Nov.2025

- Responded to patient requests and comfort needs, served meals, and restocked ward supplies such as PPE, linens and water.
- Organized and led weekly group activities for elderly patients, keeping sessions engaging and inclusive (3 hours/week).

ICEE: Road Hazard Detector | L'Odyssée de l'objet en 48 h, Hardware Lead - Team of 3

Montréal, QC | Nov.2025

- Designed and pitched a camera and laser device for buses and emergency vehicles that detects potholes, snow buildup and black ice and warns drivers, competing against ~150 students from 15 colleges in a 48-hour design competition.
- Featured in a [NeoMedia article](#) for the project concept.

Technical Skills

Mechanical: SolidWorks, Fusion 360 (7 years), AutoCAD • Stress Analysis • Additive Manufacturing • Rapid Prototyping

Control: Closed-Loop Control • Inverse Kinematics • Field-Oriented Control • PID Loops • Optical and Magnetic encoding

Programming: C++ • Python • Reinforcement Learning (MuJoCo & PyTorch / Stable-Baselines3)

Embedded: Arduino • ESP32 • SPI, UART, I2C • Soldering • System integration • Sensor Multiplexing & Matrixing

Education

University of Waterloo | Honours Mechatronics Engineering (Co-op)

Waterloo, ON | Sep.2026 – Expected Apr.2031

John Abbott College | Science Program, Dean's List

Sainte-Anne-de-Bellvue, QC | Aug.2025 – Jun.2026